Data differentiation boundary | 150 times × 9 actual receiver channels

a Observed (saved target data)

b Predicted (native starting model)

c Residual = predicted − observed

d Saved AD cotangent: L2

e Saved AD cotangent: processed

f Processed cotangent − residual

Post-update time (simulation T)

Receiver channel order (tick = node index)

Displacement (simulation units)

Objective / displacement (simulation units)

Numeric difference (simulation units) (simulation units)